

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-DC-3401A-04 Sheet 1 of 1
OPL Title	Hydrogen Reduction Safety & Exotherm Control	Discipline	Process / Operations	
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	Area / Unit	3400 - CATALYST REDUCTION SYSTEM	
Related Interlock	SEQ-3401 (SIL 2)	P&ID Ref	TJC-LLD-PID-3401	
Classification:	☐ Basic Knowledge	☐ Improvement	☒ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **hydrogen reduction safety & exotherm control** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. Hydrogen is flammable over 4-75 % in air. During reduction the PdO->Pd reaction is exothermic and the bed temperature must be watched closely. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Introduce H2 only after inerting and bed homogeneity are confirmed	Permit valid & briefed
2	Hold a constant N2+H2 flow for about 12 hours	Parameter within limit
3	Watch TE-3401 for any exotherm approaching the TSHH-3401 limit	Interlock status confirmed
4	If bed exceeds 230 degC the logic auto-trips heater and blocks H2	Change logged on DCS
5	Keep the reactor vent path (PV-3404) available at all times	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Bed multipoint TC point 3 reading open circuit	Thermocouple element failed at the sheath junction	Replaced TC point 3 in the multipoint assembly, verified reading.
One heater bank drawing no current, slow heat-up	Electric heater element bank open circuit / element burnt	Replaced heater element bank, meggered, verified TIC-17343 response
Inerting O2 analyzer drift vs portable reference	Analyzer zero drift and sample dryer saturated	Recalibrated AI-3401 on span gas, replaced sample dryer

6. KEY LEARNING POINTS

- Hydrogen is flammable over 4-75 % in air. During reduction the PdO->Pd reaction is exothermic and the bed temperature must be watched closely.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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